

Unsupervised Anomaly Detection on Attributed Networks With Graph Contrastive Learning for Consumer Electronics Security

Bo Xu¹, Member, IEEE, Jinpeng Wang², Zhehuan Zhao², Member, IEEE, Hongfei Lin²,
and Feng Xia¹, Senior Member, IEEE

Abstract—The proliferation of consumer electronic products has engendered a substantial surge in data generation and information exchange, concurrently escalating the potential for security threats. Detecting anomalies effectively on attributed networks has undeniable positive significance for consumer electronic security, such as fraudulent user detection, malicious consumption actions analysis, and network threat detection. However, the lack of real tags poses great challenges for detecting anomalies. Therefore, this paper introduces an unsupervised learning framework, namely ADVANCE, to jointly optimize the graph contrastive learning module and the network reconstruction module to accurately discover the anomalies on attributed networks in unsupervised scenarios. Specifically, ADVANCE first constructs the target view and the self-enhanced view to encode view representations by maximizing the consistency between the two views with graph contrastive learning. Subsequently, a network reconstruction module is introduced to assess the anomaly status of each node based on the degree of consistency from both topological structure and node attributes perspectives. The nodes with higher reconstruction errors are considered as anomalous nodes. Finally, the two complementary modules are jointly trained to enhance the accuracy of anomaly detection. Extensive experimental results on three benchmark datasets demonstrate the remarkable effectiveness of our proposed framework in unsupervised anomaly detection.

Index Terms—Anomaly detection, attributed networks, graph contrastive learning, unsupervised learning, consumer electronics security.

I. INTRODUCTION

WITH the popularity of consumer electronics [1], [2], a substantial volume of electronic consumption behaviors rapidly disseminate across the Internet, which constitutes a complex attributed network and fosters an increasing number

of financial fraudulent users and malicious consumer behaviors. This burgeoning landscape poses significant security challenges for consumer electronics, consequently attracting extensive research attention in recent years [3], [4], [5]. Detecting anomalies on attributed networks can facilitate the early identification of fraudulent users and financial fraud activities, which plays a crucial role in purifying the online consumption environment and ensuring the safety of consumer electronics.

Attribute networks are prevalent across various practical applications. Compared with the conventional plain networks which solely encompass interactions between nodes, attributed networks contain rich feature information of each node [6], [7], [8], providing possibilities for modeling more complex interaction systems. Detection of anomalous nodes in attributed networks is of great significance for various security-related applications, and has become an urgent research question in recent years, such as social spam detection [9], [10], financial fraud detection [11], [12], and network intrusion detection [13], [14], [15]. However, the anomalous patterns of nodes on attributed networks are related to not only the interactions between nodes but also the inconsistency of node attributes. Thus, detecting anomalies on attributed networks is highly challenging.

Early anomaly detection methods leverage shallow mechanisms such as CUR decomposition [16], subspace selection [17], ego-network analysis [18], and residual analysis [19], [20] to identify anomalous nodes. However, these methods usually rely on feature engineering or observed node interactions, which exhibit insensitivity to the nonlinear characteristics of networks. Recently, graph neural networks have found widespread application in anomaly detection [21], [22], [23], [24], which aim to capture network nonlinearity and learn node representations for anomalous patterns with deep neural networks. Due to the high cost and unreliability associated with labeling anomalies, unsupervised methods for anomaly detection are more reliable [25], [26].

The rise of graph contrastive learning provides a new direction for unsupervised anomaly detection [28], [38]. Graph contrastive learning has promising properties for anomaly detection. *First*, maximizing mutual information strategy of graph contrastive learning can effectively learn the deep information of graphs without the guidance of labels. *Second*, graph contrastive learning models measure the consistency

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Bo Xu, Jinpeng Wang, and Zhehuan Zhao are with the School of Software, Dalian University of Technology, Dalian 116620, China (e-mail: boxu@dlut.edu.cn; wangjinpeng@dlut.edu.cn; z.zhao@dlut.edu.cn).

Hongfei Lin is with the School of Computer Science and Technology, Dalian University of Technology, Dalian 116620, China (e-mail: hflin@dlut.edu.cn).

Feng Xia is with the School of Computing Technologies, RMIT University, Melbourne, VIC 3000, Australia (e-mail: f.xia@ieee.org).

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among elements within each instance pair and the consistency is highly correlated with the abnormality of instance. However, existing SOTA (state-of-the-art) methods based on contrastive learning typically constructed positive and negative instance pairs based on node subgraphs. These methods learned node representations by maximizing the mutual information (MI) between the target node and positive instances and minimizing the mutual information (MI) between the target node and negative instances, which may not be effective for learning heterogeneous graphs, where the target node and its neighbors do not belong to the same category.

To overcome the shortcomings, some studies have constructed different view information to learn the embedding representations of networks by maximizing the similarity between different views [29], [56]. Based on this inspiration, we can construct views with different structural information and encode the node representations by maximizing the MI of different views, so as to detect anomalies more efficiently and ensure the safety of consumer electronics.

In this paper, we propose a novel view-level unsupervised contrastive learning framework for anomaly detection on attributed networks (ADVANCE for abbreviation), which contains a graph contrastive learning module and a network reconstruction module. The graph contrastive learning module aims to maximize the consistency between the target view and the self-enhanced view. The network reconstruction module measures the reconstruction error of nodes from both the structural and attribute perspectives. Specifically, ADVANCE first generates a target view with edge attributes information from the original data as the learning guidance, and then constructs a self-enhanced view by using the full graph parameterization method and k -nearest neighbors algorithm. Next, two weight-shared graph convolutional networks are used to generate representations of the two views. After obtaining the representations of the two views, we construct a contrastive loss to maximize the MI between the target view and the self-enhanced view, thereby encoding deeper potential information of the graph. The network reconstruction module receives the potential representations of the self-enhanced view and reconstructs the original network from two perspectives: node attributes and network structure. Finally, the two complementary modules are integrated to detect anomalous nodes through reconstruction errors of nodes. Our primary contributions can be summarized as follows:

- We design a view-level contrastive learning model that encodes view representations by maximizing the consistency between the self-enhanced view and the well-designed target view with edge attributes.
- We propose a novel unsupervised learning framework, namely ADVANCE, for anomaly detection on attributed networks. ADVANCE can effectively integrate graph contrastive learning-based and network reconstruction-based modules and enhance the efficiency of anomaly detection by jointly optimizing the two complementary modules, which provides a high guarantee for the safety of consumer electronics.
- We perform extensive experiments on three benchmark datasets to evaluate the performance of the ADVANCE.

The results validate the effectiveness of ADVANCE, which outperforms existing SOTA methods.

The rest of the paper is organised as follows: Section II summarizes related work. Section III introduces the concept of attributed networks and formally formulates the problem of unsupervised anomaly detection on attributed networks. Section IV discusses our optimal solution (ADVANCE) and describes its architecture in detail, and Section V represents the experiments and results. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Anomaly Detection on Attributed Networks

Recently, how to detect anomalies on attributed networks has aroused extensive research interest. Anomaly detection can be regarded as a data mining process that aims to identify data points deviating from the normal patterns within a dataset [30], [31]. Early anomaly detection works relied heavily on feature engineering constructed by domain experts [32], [33], which not only requires a lot of domain knowledge but also limits the performance of anomaly detection methods. Many machine learning techniques are also used to detect anomalies on attributed networks, Radar [19] characterized the residuals of attributes information and its coherence with network information, addressing the issue of complicating the residual modeling process due to inter-instance interactions. ANOMALOUS [16] combined CUR decomposition and residual analysis, treating them as a whole for attribute selection and anomaly detection. However, these shallow methods cannot effectively be extended to high-dimensional data and lack the ability to capture non-linear attributes in networks.

To alleviate the above drawbacks, recent studies have adopted deep learning techniques to detect anomalies on attributed networks [25], [34]. Dual-SVDAE [36] introduced a dual hypersphere learning mechanism to learn hyperspheres for normal nodes and measures the distances from nodes to the learning centers of these hyperspheres to detect anomalous nodes. CFAD [35] learned causal relations in attributed networks and identified abnormal nodes from the perspective of structure and context. GraphBEAN [37] focuses on the rich interactions in bipartite graphs, which uses a bipartite node-and-edge-attributed graph to generate anomaly scores for each edge and each bipartite node, respectively. Although these methods have achieved better performance than shallow approaches by using deep neural networks, the single graph convolutional network encoder still has the limitation of not being able to fully model the complex attributes of the network. Furthermore, marking anomalous nodes is also a challenging task.

With the popularization of contrastive learning, the application of this approach to anomaly detection on attributed networks has attracted the attention of researchers. Sub-CR [38] established two contrastive learning views to encode local and global information related to anomalies, and then introduced a reconstruction module based on masked autoencoder to identify nodes with large reconstruction errors as

anomalous nodes. Hu and Shao [39] proposed an improved unsupervised contrastive learning method, which comprehensively compared both the internal and external aspects of subgraphs and leverages a trained teacher model as prior knowledge to modify the sampling probabilities for selectively aggregating neighbor nodes. However, related researches in this field remain limited, and the development of new anomaly detection methods based on contrastive learning is still worth exploring.

B. Graph Contrastive Learning

Contrastive learning models the general characteristics of networks by letting the model learn which nodes are similar or different without labels. The great success of contrastive learning in computer vision [40], [41] has motivated its continuous migration to the field of graph learning. DGI [57] maximized the mutual information between graph-enhanced representations and the currently extracted graph information, enabling the node representations learning in an unsupervised setting. GMI [58] proposed a novel concept, Graphical Mutual Information (GMI), which is trained by maximizing the GMI between the input and output of a Graph Neural Encoder. GIC [42] performed joint optimized clustering of nodes and retained Infomax target items of MI between nodes of the same clustering. The fundamental idea behind graph contrastive learning is to maximize the mutual information between nodes, which brings the representations of semantically similar nodes closer while simultaneously distancing the representations of semantically unrelated nodes. However, the above-mentioned methods aim to maximize the MI between the embeddings of the target node and its neighboring nodes, which may not be effective for learning heterogeneous graphs, where the target node and its neighbors do not belong to the same category.

Recent studies have considered the above problems and constructed different view information to learn the representations of graphs by maximizing the similarity between different views. MVGRL [29] proposed a contrastive multi-view representation learning method by contrasting embeddings from two structural views of graphs. SAIL [43] performed self-augmented graph contrastive learning through two complementary self-distilling regularization modules, i.e., intra-graph and inter-graph knowledge distillation. SUBLIME [44] constructed an “anchor graph” from the original data as a learning guide, and utilized contrastive learning to maximize the mutual information between the anchor graph and the learned graph. Graph contrastive learning is also beneficial for diverse applications, such as chemical prediction [45], [46], federated learning [47], [48], and recommendation [49], [50], [51]. Numerous studies have demonstrated the wide application of graph contrastive learning, but how to detect anomalies on attributed networks with graph contrastive learning is still a novel and challenging research topic.

III. PROBLEM FORMULATION

In this section, we formally define attributed networks and the unsupervised anomaly detection problem on attributed

TABLE I
DESCRIPTION OF NOTATIONS

Notation	Explanations
N/d	The number of nodes/features
$X \in \mathbb{R}^{N \times d}$	The initial attribute matrix for nodes
$A \in \mathbb{R}^{N \times N}$	The original graph adjacency matrix
$A_t \in \mathbb{R}^{N \times N}$	The target view adjacency matrix
$S \in \mathbb{R}^{N \times N}$	The self-enhanced view adjacency matrix
$\mathcal{G}_t = (A_t, X)$	The target view
$\mathcal{G}_s = (S, X)$	The self-enhanced view
λ^{EA}	The attributes matrix of edges
$\mathcal{L}_{cl}$	The contrastive loss between $\mathcal{G}_t$ and $\mathcal{G}_s$
$\mathcal{L}_{res}$	The reconstruction loss of network structure and node attributes
$\mathcal{L}$	The total loss of ADVANCE framework
$N(v_i)$	The neighbors set of node v_i
$\tilde{A} \in \mathbb{R}^{N \times N}$	The reconstructed adjacency matrix
$\tilde{X} \in \mathbb{R}^{N \times N}$	The reconstructed attributes matrix for nodes
$F_\omega(\cdot)$	The full graph parameterization learner with parameter ω
$sym(\cdot)/nor(\cdot)$	The symmetrization and normalization functions
$sim(\cdot)$	The similarity function
$score(\cdot)$	The anomaly score function
$(\cdot)^T$	The transposition operation
k	The number of nearest neighbors
ϑ	The temperature factor
α	The control parameter for balancing the influence of structure reconstruction and attribute reconstruction
μ	The tuning parameter that measures the importance of graph contrastive learning module and network reconstruction module
δ	The structure feedback mechanism decay rate
ε	The structure feedback mechanism interval

networks. Table I lists the description of the notations mainly used in this paper.

Definition 1 (Attributed Networks): An attributed network can be represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X)$, where $\mathcal{V} = \{v_1, \dots, v_N\}$ and $\mathcal{E}$ denote the set of nodes and edges, respectively. $X \in \mathbb{R}^{N \times d}$ is the attribute matrix, where N is the number of nodes in the attributed network and d is the dimension of attributes. The i^{th} row vector $x_i \in \mathbb{R}^d$ in the attribute matrix is the attributes for the i^{th} node v_i .

Definition 2 (Unsupervised Anomaly Detection on Attributed Networks): The objective of this paper is to detect nodes with significant differences in attributed networks based on structure and attribute vectors in an unsupervised setting. Specifically, given an attributed network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X)$, ADVANCE learns an anomaly score function $score(\cdot)$ to calculate each node's anomaly score $score(v_i)$, which measures the degree of abnormality for each node without anomaly labels as training guidance. Generally, nodes with higher anomaly scores are more likely to be considered as anomalous nodes.

IV. THE FRAMEWORK OF ADVANCE

This section elaborates on our proposed ADVANCE, which is shown in Fig. 1. The following subsections provide a detailed overview of each module respectively.

A. Graph Contrastive Learning Module

As a critical component of ADVANCE, Graph Contrastive Learning Module aims to establish two distinct views to

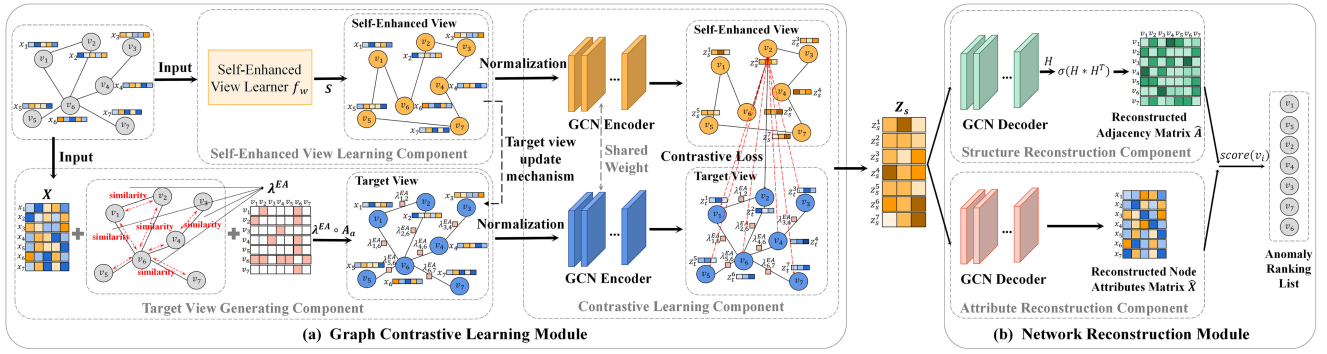


Fig. 1. The overall pipeline of ADVANCE. The target is to predict the anomaly score of each node in attributed networks. In graph contrastive learning module (a), “node v.s. node” consistency is learned by contrasting the target view with the self-enhanced view, thereby encoding deep information of the graph. In network reconstruction module (b), the optimal graph representation encoded by the contrastive learning module is used to reconstruct node attributes and topological structure, and to identify anomalous nodes through reconstruction loss.

TABLE II
CALCULATION METHODS FOR SIX SIMILARITY INDICES

Indices	Calculation Methods
$d(v_i)$	The number of edges connected to node v_i
$d(N(v_i))$	The number of edges connected to node v_i 's neighbors
$CN(v_i, v_j)$	$CN(v_i, v_j) = N(v_i) \cap N(v_j) $
$Adamic(v_i, v_j)$	$Adamic(v_i, v_j) = \sum_{k \in N(v_i) \cap N(v_j)} \frac{1}{\log N(k) }$
$Jaccard(v_i, v_j)$	$Jaccard(v_i, v_j) = \frac{ N(v_i) \cap N(v_j) }{ N(v_i) \cup N(v_j) }$
$PA(v_i, v_j)$	$PA(v_i, v_j) = d(v_i) \times d(v_j)$

contrast: the self-enhanced view that learns the optimal graph latent representations and the target view that guides representation learning. Subsequently, node-level contrastive learning is employed to maximize consistency between the two views, thus encoding graph information effectively.

1) *Target View Generating Component*: A target view is defined as a view constructed directly from input data rather than being learned, which aims to provide correct guidance for graph contrastive learning. In this paper, we fully consider the interaction information between nodes and attributes information contained in the attributed network and construct the target view $\mathcal{G}_t = (A_t, X)$ with edge attributes information, where $A_t \in \mathbb{R}^{N \times N}$ denotes the adjacency matrix with edge attributes, $X \in \mathbb{R}^{N \times d}$ denotes the attribute matrix for nodes. Specifically, six similarity indices are selected to calculate the similarity of two connected nodes as edge attributes, including the degree of node v_i , the degree of node v_i 's neighbors, common neighbors (CN), Admaic-Adar Index, Jaccard Index, and Preferential Attachment Index (PA). TABLE II lists the calculation methods for six similarity indices, where the $N(v_i)$ denotes the neighbor's set of node v_i . Next, the adjacency matrix A_t with edge attributes is constructed as:

$$A_t = \lambda^{EA} \circ A \quad (1)$$

where λ^{EA} is the edge attribute matrix calculated by the similarity of two connected nodes, A denotes the original graph adjacency matrix and $\circ$ represents the Hadamard product. To ensure the stability and effectiveness of the learning process, we implement a target view update mechanism to update the target view, which will be described in Section IV-A4.

2) *Self-Enhanced View Learning Component*: The main function of the Self-Enhanced View Learning Component is to generate a self-enhanced adjacency matrix $S \in \mathbb{R}^{N \times N}$ with a parameterized model to construct self-enhanced view $\mathcal{G}_s = (S, X)$ for graph contrastive learning. Inspired by previous work [44], we use the full graph parameterization method as the self-enhanced view learner:

$$S = F_\omega(X) = \sigma(W) \quad (2)$$

where $\omega = W$ is a parameter matrix and σ is a non-linear activation function. Full graph parameterization method assumes that each edge exists independently in the graph, so the adjacency matrix can be optimized directly without the participation of node features, which is very advantageous for modeling attributed networks with node attribute anomalies.

However, the adjacency matrix S constructed by the full graph parameterization method is usually dense, which will mask important interactions of the network and consume computing resources extremely. Hence, we use the k -nearest neighbors (knn) algorithm to reserve the edges with top- k connection values for each node as its neighborhood structure to build the sparse self-enhanced adjacency matrix S :

$$S_{ij} = \begin{cases} S_{ij}, & S_{ij} \in \text{top}_k(S_i) \\ 0, & S_{ij} \notin \text{top}_k(S_i) \end{cases} \quad (3)$$

where $\text{top}_k(S_i)$ is the set of top- k values of row vector S_i .

3) *Contrastive Learning Component*: Given the target view and self-enhanced view, we perform the normalization process to ensure that the adjacency matrix is symmetric and consists of non-negative elements:

$$A = \text{sym}(\text{nor}(A)) = \frac{\sigma(A) + \sigma(A)^T}{2} \quad (4)$$

where σ is a non-linear activation function that maps elements to $[0, 1]$ and $(\cdot)^T$ denotes the transposition operation.

After obtaining normalized views (i.e., $\mathcal{G}_t$ and $\mathcal{G}_s$), two weight-shared Graph Convolutional Network Encoders [52] are utilized to generate multi-view representations. Specifically, we input the target view $\mathcal{G}_t$ and the self-enhanced view $\mathcal{G}_s$ into GCN1 and GCN2 to perform neighborhood

information aggregation and output corresponding view representations Z_t and Z_s :

$$\begin{cases} Z_t = GCN1(\mathcal{G}_t) = MLP\left(\sigma\left(\hat{A}_t\sigma\left(\hat{A}_tXW^{(0)}\right)W^{(1)}\right)W^{(2)}\right) \\ Z_s = GCN2(\mathcal{G}_s) = MLP\left(\sigma\left(\hat{S}\sigma\left(\hat{S}XW^{(0)}\right)W^{(1)}\right)W^{(2)}\right) \end{cases} \quad (5)$$

where $\hat{A}_t = D_t^{-1/2}(A_t + I)D_t^{-1/2}$, $\hat{S} = D_s^{-1/2}(S + I)D_s^{-1/2}$, $X \in \mathbb{R}^{N \times d}$ represents the attribute matrix of nodes, $W^{(0)}$, $W^{(1)}$, $W^{(2)}$ is the learnable weight matrix in Graph Convolutional Network Encoders. Here, D_t and D_s are the diagonal degree matrix of $(A_t + I)$ and $(S + I)$, respectively. We notice that using only A means that only the characteristic information of all nodes' neighbors is considered, but the nodes themselves are neglected. Therefore, we add self-connections in the graph (i.e., using $(A + I)$) to ensure that the nodes' own characteristic information also participates in aggregation.

The main objective of contrastive learning is to encourage the representations of semantically similar pairs (x, x^+) to be close, while the representations of semantically dissimilar pairs (x, x^-) should be pushed apart. To achieve the above goal, a contrastive loss is designed to maximize the similarity between the same nodes' representations from different views. Specifically, in Eq. (6) and Eq. (7), after obtaining the representations Z_t and Z_s , we apply a symmetric normalized temperature-scaled cross-entropy loss to design a node-level contrastive loss function for the target view and self-enhanced view, which can pull the representations of the same nodes from different views close to each other while pushing the representation of one node far away from the representations of other nodes from both the same view and different views:

$$\mathcal{L}_{cl}^1(v_i) = -\log \frac{e^{\text{sim}(Z_t^{v_i}, Z_s^{v_i})/\vartheta}}{\sum_{j=1}^N e^{\text{sim}(Z_t^{v_i}, Z_s^{v_j})/\vartheta}} \quad (6)$$

$$\mathcal{L}_{cl}^2(v_i) = -\log \frac{e^{\text{sim}(Z_s^{v_i}, Z_t^{v_i})/\vartheta}}{\sum_{j=1}^N e^{\text{sim}(Z_s^{v_i}, Z_t^{v_j})/\vartheta}} \quad (7)$$

where ϑ represents the temperature factor that controls the concentration level of the distribution, $\text{sim}(\cdot)$ is the similarity function to measure the similarity of representations of two nodes. Combining the above two losses, the objective function of the graph contrastive learning module is defined as follows:

$$\mathcal{L}_{cl} = \frac{1}{2N} \sum_{i=1}^N (\mathcal{L}_{cl}^1(v_i) + \mathcal{L}_{cl}^2(v_i)) \quad (8)$$

where N is the number of nodes in the attributed network. Through the above equation, we can encode graph information and learn the optimal graph representation without the guidance of labels.

4) *Target View Update Mechanism*: In order to make the self-enhanced view learn the graph representations containing fewer anomalous information, we design a target view update mechanism during the contrastive learning process. The main idea of the target view update mechanism is to update the target structure slowly according to the probability of edges existing between every pair of nodes calculated by

the constantly learning representations of self-enhanced view Z_s instead of keeping the target view $\mathcal{G}_t$ unchanged. The assumption behind the target view update mechanism is that if there is a real link between two nodes, its connectivity patterns can be well reconstructed. On the contrary, if the probability of a connection existing between two nodes is low, it implies that there is no link or anomalous link between them. Specifically, we take the latent representations of self-enhanced view Z_s as input and predict whether there is a link between each pair of two nodes:

$$P = \text{sigmoid}(Z_s Z_s^T) \quad (9)$$

Next, given a decay rate $\delta \in [0, 1]$, the target view structure A_t is updated every ε iterations as follows:

$$A_t = \delta A_t + (1 - \delta)P \quad (10)$$

Through the target view update mechanism, the self-enhanced view $\mathcal{G}_s$ can constantly correct anomalous patterns existing in the target view and alleviate structural anomaly.

B. Network Reconstruction Module

The network reconstruction module consists of structure reconstruction component and attribute reconstruction component, with the objective of decoding the self-enhanced view representations Z_s to reconstruct network structure information and node attributes information. After that, anomalous nodes can be identified by the reconstruction error of nodes.

1) *Structure Reconstruction Component*: The structure reconstruction component mainly discusses how to reconstruct the original topological structure with the learned potential representations Z_s from the graph contrastive learning module. If $\tilde{A}$ denotes the reconstructed adjacency matrix, then the structural anomalies on the network can be determined by using the structure reconstruction error $R_s = \|A - \tilde{A}\|_F^2$. Specifically, if the neighborhood structural information of a certain node can be effectively approximated to the original structure through the structure reconstruction component, then it is less likely that the node is an anomalous node. On the contrary, if the neighborhood structural information of a certain node cannot be effectively reconstructed, it implies the presence of abnormal patterns in its structural information. Therefore, a larger norm of $R_s^i = \|a_i - \tilde{a}_i\|_F^2$ indicates a higher probability of network structural anomalies for the i -th node on the attributed network. Particularly, we first use a graph convolutional network as a structure reconstruction decoder to decode the input potential representation Z_s :

$$H = GCN3(Z_s, A) = MLP\left(\sigma\left(\hat{A}_s Z_s W^{(3)}\right)W^{(4)}\right) \quad (11)$$

where $W^{(3)}$, $W^{(4)}$ is the learnable weight matrix in the structure reconstruction decoder. Then, we predict whether there is a link between each pair of nodes based on the decoding vector H to generate the reconstructed adjacency matrix $\tilde{A}$:

$$\tilde{A} = \text{sigmoid}(HH^T) \quad (12)$$

2) *Attribute Reconstruction Component*: In order to calculate the attribute reconstruction error $R_A = \|X - \tilde{X}\|_F^2$ to determine the attributed anomalies on the network, we propose an attribute reconstruction decoder to approximate the original node attributes information according to the potential representation Z_s generated by the graph contrastive learning module. Similarly, if the reconstructed attributes of a node are closer to the original attributes, the node is more inclined to be a normal node. On the opposite side, if the reconstructed attributes of a node are quite different from the original attributes, it is of high probability to be anomalous. Specifically, we utilize another graph convolutional network as the attribute reconstruction decoder to reconstruct the original attributes of nodes as follows:

$$\tilde{X} = GCN4(Z_s, A) = MLP\left(\sigma\left(\hat{A}\sigma\left(\hat{A}Z_s W^{(5)}\right)W^{(6)}\right)W^{(7)}\right) \quad (13)$$

where $W^{(5)}, W^{(6)}, W^{(7)}$ is the learnable weight matrix. To jointly learn the reconstruction errors, we combine the structure reconstruction component and the attribute reconstruction component to construct the objective function of the network reconstruction module:

$$\mathcal{L}_{res} = \alpha R_A + (1 - \alpha) R_S = \alpha \|X - \tilde{X}\|_F^2 + (1 - \alpha) \|A - \tilde{A}\|_F^2 \quad (14)$$

where α is an important control parameter to balance the influence of structure and attribute reconstruction.

C. Anomaly Detection

To jointly train the graph contrastive learning module and the network reconstruction module, we optimize the following objective function:

$$\mathcal{L} = \mu \mathcal{L}_{cl} + (1 - \mu) \mathcal{L}_{res} \quad (15)$$

where μ is a non-negative tuning parameter that measures the importance of two complementary modules. Then an anomaly score function is defined to detect anomalies. Inspired by [25], taking node v_i as an example, we can calculate the anomaly score of node v_i according to the reconstruction error:

$$score(v_i) = \alpha \|x_i - \tilde{x}_i\|_2 + (1 - \alpha) \|a_i - \tilde{a}_i\|_2 \quad (16)$$

where $\tilde{x}_i(x_i)$ is the reconstructed (original) attributes vector of the node v_i , $\tilde{a}_i(a_i)$ is the i -th row vector of the reconstructed adjacency matrix $\tilde{A}$ (original graph adjacency matrix A), which denotes the reconstructed (original) structure vector of node v_i . Specially, nodes with higher scores are more likely to exhibit anomalous patterns, thus we can sort the nodes according to the anomaly scores to identify anomalous nodes. The calculation process of the ADVANCE framework is summarized as Algorithm 1.

V. EXPERIMENTS

In this section, we first introduce the dataset, the comparison algorithm, and the evaluation criteria used in the experiments. Next, we conduct numerous experiments to substantiate the superior validity of the proposed framework ADVANCE. We

Algorithm 1: The Calculation Process of ADVANCE

Input: Node attributes matrix X ; Original graph adjacency matrix A ; Number of nearest neighbors k ; Control parameters for balancing the structure reconstruction and attribute reconstruction α ; Tuning parameters for balancing graph contrastive learning module and network reconstruction module μ ; Temperature factor ϑ ; Target view update mechanism decay rate δ ; Target view update mechanism interval ε ; Epoch number ξ ;

Output: Anomaly score of all nodes $score(\mathcal{V})$;

- 1 Calculate edge attribute matrix λ^{EA} by the similarity of two connected nodes;
- 2 Initialize target view $\mathcal{G}_t = (A_t, X)$ by $A_t = \lambda^{EA} \circ A$;
- 3 Initialize parameters $\{\omega, W^{(0)}, W^{(1)}, W^{(2)}, W^{(3)}, W^{(4)}, W^{(5)}, W^{(6)}, W^{(7)}\}$;
- 4 **for** $iter = 1$ to ξ **do**
- 5 Calculate S with Self-Enhanced View Learning Component;
- 6 Establish self-enhanced view $\mathcal{G}_s = (S, X)$;
- 7 Normalize $\mathcal{G}_t, \mathcal{G}_s$ with normalization operation;
- 8 Calculate view representations Z_t, Z_s with graph contrastive learning encoder;
- 9 Calculate graph contrastive loss $\mathcal{L}_{cl}$;
- 10 Calculate reconstructed adjacency matrix $\tilde{A}$ with Structure Reconstruction Component;
- 11 Calculate reconstructed attribute matrix $\tilde{X}$ with Attribute Reconstruction Component;
- 12 Calculate network reconstruction loss $\mathcal{L}_{res}$;
- 13 Calculate total loss of ADVANCE framework $\mathcal{L}$;
- 14 Update parameters $\{\omega, W^{(0)}, W^{(1)}, W^{(2)}, W^{(3)}, W^{(4)}, W^{(5)}, W^{(6)}, W^{(7)}\}$ via back propagation;
- 15 **if** $iter \bmod \varepsilon = 0$ **then**
- 16 Calculate the probability P that there is a link between each pair of nodes according to Z_s ;
- 17 Update $\mathcal{G}_t$ by target view update mechanism decay rate δ ;
- 18 **end**
- 19 **end**
- 20 Calculate the anomaly score of all nodes $score(\mathcal{V})$ according to the reconstruction error;

aim to answer three research questions as follows: **RQ1**: How effective is ADVANCE for detecting anomalies on attributed networks under unsupervised settings? **RQ2**: How do key hyper-parameters influence the performance of ADVANCE? and **RQ3**: How do the contrastive learning component, structure reconstruction component, and attribute reconstruction component impact the performance of ADVANCE?

A. Datasets

To assess the performance of ADVANCE in anomaly detection, we employed three real-world attributed network datasets that have been widely utilized in previous studies [28], [53]:

TABLE III
THE STATISTICS OF THE DATASETS

Dataset	#nodes	#edges	#attributes	#anomalies
Cora	2708	5429	1433	150
BlogCatalog	5196	171743	8189	300
Flickr	7575	239738	12047	450

- **Cora:** Cora is an attributed network that contains seven categories of published papers related to machine learning. It takes the form of a citation network where each paper serves as a node, and edges denote the citation relations between different papers.
- **BlogCatalog:** BlogCatalog constructs a social network by gathering data from the blog-sharing website BlogCatalog, where the users are represented as nodes, and edges symbolize the following relationships between users.
- **Flickr:** Flickr serves as a website for image management and sharing. Similar to BlogCatalog, users on Flickr have the ability to mutually follow each other, thereby forming a social network.

Given the absence of ground truth anomalies in the aforementioned datasets, it is necessary to inject anomalies into a clean attributed network to evaluate our proposed framework. Specifically, we refer to two anomaly injection methods commonly employed in previous researches [25], [54] to inject structure anomaly and attribute anomaly for each dataset. The statistics of the datasets are presented in TABLE III.

B. Baselines and Evaluation Criteria

We conduct a comparative analysis of ADVANCE against nine mainstream anomaly detection methods:

- **Radar** [19]: Radar models the residuals of node attributes and their coherence with network structure, serving the purpose of anomaly detection.
- **ANOMALOUS** [16]: ANOMALOUS jointly optimizes attribute selection and anomaly detection using CUR decomposition and residual analysis methods.
- **DOMINANT** [25]: DOMINANT utilizes graph convolutional autoencoder to jointly reconstruct network structure and node attributes to identify anomalous nodes.
- **DGI** [57]: DGI maximized mutual information between patch representations and corresponding high-level graph summaries to characterize network information.
- **DeepAE** [55]: DeepAE uses Laplacian sharpening to enlarge the difference between normal and anomalous nodes and calculate reconstruction errors to detect anomalies.
- **Deep2NAD** [26]: Deep2NAD utilizes a Variational AutoEncoder (VAE) to detect anomalies by simultaneously considering both network structure and node attributes.
- **ANEMONE** [27]: ANEMONE performs multi-scale contrastive learning at both patch and context levels for anomaly detection.

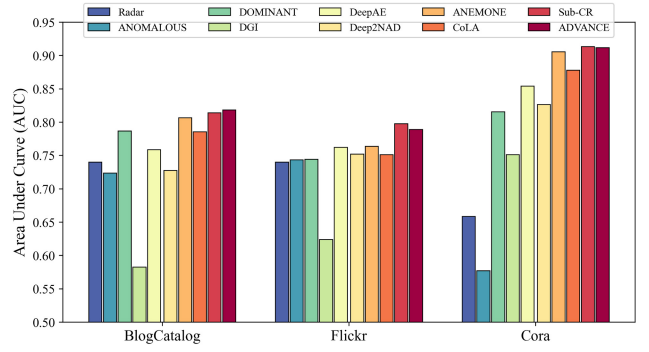


Fig. 2. Performance of different anomaly detection methods w.r.t. AUC.

- **CoLA** [28]: CoLA utilizes contrastive learning to learn representations from node-subgraph instance pairs and provides the discriminant scores of anomaly ranking.
- **Sub-CR** [38]: Sub-CR utilizes multi-view contrastive learning to integrate local and global information so as to provide high features for detecting anomalous nodes.

Furthermore, we use the average of three evaluation metrics (i.e., AUC, Precision@K, and Recall@K) obtained from 5 independent training cycles to measure the performance of the ADVANCE framework and baseline. The details of these three evaluation metrics are extensively elaborated in [25].

C. Experiment Reproducibility

For the proposed framework ADVANCE, the number of epoch and learning rate are set as 3000 and 0.005 for BlogCatalog dataset and Flickr dataset and Cora dataset. To generate the target view, the common neighbors (CN) is selected as the edge attributes for the Flickr dataset and the average of the calculated results of six similarity indices as the edge attributes for the remaining datasets. For constructing self-enhanced view, we use full graph parameterization learner on all three datasets, and set the number of nearest neighbors k to 40, 30, and 10 for the BlogCatalog dataset, Flickr dataset, and Cora dataset, respectively. The tuning parameter μ that measures the importance of graph contrastive learning module and network reconstruction module are set to 0.8, 0.8, and 0.9 for the BlogCatalog dataset, Flickr dataset, and Cora dataset, respectively. For the control parameter α that balances the influence of structure reconstruction and attribute reconstruction, we follow the settings of previous work [25], [27]. For the BlogCatalog and Flickr datasets, a reasonable choice for α is approximately 0.4 to 0.7, while for the Cora dataset, a reasonable choice is around 0.7 to 0.8. Finally, for the decay rate δ and interval ε in the structure feedback mechanism, the most suitable combination of parameters is selected for each dataset according to Section V-E, which are $\{\delta = 0.9999, \varepsilon = 0\}$, $\{\delta = 0.9999, \varepsilon = 0\}$, and $\{\delta = 0.999, \varepsilon = 0\}$ for BlogCatalog dataset, Flickr dataset, and Cora dataset, respectively.

D. Performance Comparison (RQ1)

Fig. 2 illustrates the comparison results of these methods in terms of AUC value on three datasets and TABLE IV

TABLE IV
PERFORMANCE OF DIFFERENT ANOMALY DETECTION METHODS W.R.T. PRECISION@K AND RECALL@K

Precision@K												
Datasets	BlogCatalog				Flickr				Cora			
K	50	100	200	300	50	100	200	300	50	100	200	300
Radar(2017)	0.660	0.670	0.550	0.416	0.740	0.700	0.635	0.503	0.630	0.620	0.590	0.440
ANOMALOUS(2018)	0.640	0.650	0.515	0.417	0.790	0.710	0.650	0.510	0.540	0.570	0.535	0.413
DOMINANT(2019)	0.760	0.710	0.590	0.467	0.770	0.730	0.685	0.593	0.760	0.730	0.640	0.470
DGI(2019)	0.500	0.470	0.410	0.333	0.540	0.530	0.465	0.360	0.700	0.670	0.615	0.443
DeepAE(2020)	0.680	0.670	0.545	0.427	0.740	0.710	0.655	0.547	0.780	0.750	0.665	0.470
Deep2NAD(2021)	0.600	0.640	0.525	0.410	0.760	0.780	0.665	0.583	0.800	0.770	0.670	0.467
ANEMONE(2021)	0.800	0.730	0.610	0.477	0.790	0.780	0.675	0.593	0.860	0.800	0.690	0.488
CoLA(2022)	0.760	0.720	0.600	0.470	0.780	0.770	0.680	0.590	0.810	0.780	0.685	0.476
Sub-CR(2022)	0.830	0.750	0.610	0.490	0.840	0.800	0.730	0.605	0.880	0.810	0.705	0.497
ADVANCE(Ours)	0.840	0.758	0.612	0.493	0.812	0.800	0.693	0.607	0.900	0.802	0.693	0.483

Recall@K												
Datasets	BlogCatalog				Flickr				Cora			
K	50	100	200	300	50	100	200	300	50	100	200	300
Radar(2017)	0.110	0.223	0.367	0.416	0.082	0.156	0.282	0.336	0.210	0.413	0.786	0.880
ANOMALOUS(2018)	0.107	0.217	0.343	0.417	0.087	0.158	0.289	0.340	0.180	0.380	0.713	0.827
DOMINANT(2019)	0.127	0.237	0.393	0.467	0.084	0.162	0.304	0.396	0.253	0.487	0.853	0.940
DGI(2019)	0.083	0.157	0.273	0.333	0.060	0.118	0.207	0.240	0.233	0.447	0.820	0.886
DeepAE(2020)	0.113	0.223	0.363	0.427	0.082	0.158	0.291	0.365	0.260	0.500	0.887	0.940
Deep2NAD(2021)	0.100	0.213	0.350	0.410	0.084	0.173	0.295	0.388	0.267	0.513	0.893	0.934
ANEMONE(2021)	0.133	0.243	0.407	0.477	0.088	0.173	0.300	0.395	0.286	0.533	0.920	0.976
CoLA(2022)	0.127	0.240	0.400	0.470	0.087	0.171	0.302	0.393	0.270	0.520	0.913	0.953
Sub-CR(2022)	0.138	0.250	0.407	0.490	0.093	0.178	0.324	0.403	0.293	0.540	0.940	0.993
ADVANCE(Ours)	0.140	0.253	0.408	0.493	0.090	0.178	0.308	0.405	0.300	0.534	0.924	0.967

shows the performance of our framework and baselines in terms of Precision@K and Recall@K for $K = 50, 100, 200, 300$. From the comparison results, we summarize the following conclusions: (1) Our proposed ADVANCE outperforms all baselines on the BlogCatalog dataset, and achieves the runner-up position on the Flickr and Cora datasets. The superior performance of ADVANCE verifies the effectiveness of the joint optimizing graph contrastive learning module and network reconstruction module, which can better capture node attributes and topology structure of attributed networks for anomaly detection. (2) Compared with the deep learning-based methods, shallow methods such as Radar and ANOMALOUS can not achieve satisfactory results because these methods lack the ability to model high-dimensional attributes vectors and intricate network structures. Deep learning-based methods (DOMINANT, DeepAE, and Deep2NAD) break through the limitation of shallow mechanisms to capture the highly non-linear attributes, which extensively leverages the intrinsic information of the network and achieves superior performance than the shallow methods. (3) The performance of methods based on contrastive learning (ANEMONE, COLA, Sub-CR, and ADVANCE) is superior to the shallow and deep learning methods, which indicates that the idea of mining hidden information from graph data by contrastive learning to provide auxiliary guidance for anomaly detection is feasible. However, DGI, which also adopts contrastive learning mechanism, does not show competitive performance. The reason behind this limitation lies in the use of node-to-global-instance comparisons in contrastive learning, which cannot capture anomalies within local substructure and is not friendly to heterogeneous graphs. The outstanding performance of Sub-CR can be attributed to its adept aggregation of local-global structural information, an aspect that ADVANCE lacks in its consideration. Node anomalies in the graph often

occur on different scales. For example, in social networks, a user's behavior may be considered normal within their circle of friends (i.e., local anomalies) but not across the entire network (i.e., global anomalies). Therefore, integrating local and global structural information can better capture structural features of different scales associated with anomaly patterns.

E. Parameter Analysis (RQ2)

In this subsection, we investigate the impacts of four important parameters on the performance of ADVANCE: the tuning parameter μ , the number of nearest neighbors k , the target view update mechanism decay rate δ , and the edge attributes.

Tuning parameter μ aims to balance the importance of graph contrastive learning module and network reconstruction module. When $\mu = 1$, we only use the information of graph contrastive learning module to train ADVANCE while ignoring the network reconstruction error. When $\mu = 0$, we use graph convolutional networks to encode view representations instead of using a contrastive learning mechanism to mine the intrinsic information of graphs. Fig. 3(a) shows the effect of different μ values on the performance. The performance of ADVANCE degrades either without training graph contrastive learning module ($\mu = 0$) or without training network reconstruction module ($\mu = 1$), which demonstrates that the joint optimization of two complementary modules is helpful to improve the performance of ADVANCE in detecting anomalies, especially the network reconstruction module. Furthermore, the importance of different modules varies on different datasets, and the optimal tuning parameters for BlogCatalog, Flickr, and Cora datasets are 0.8, 0.8, and 0.9, respectively.

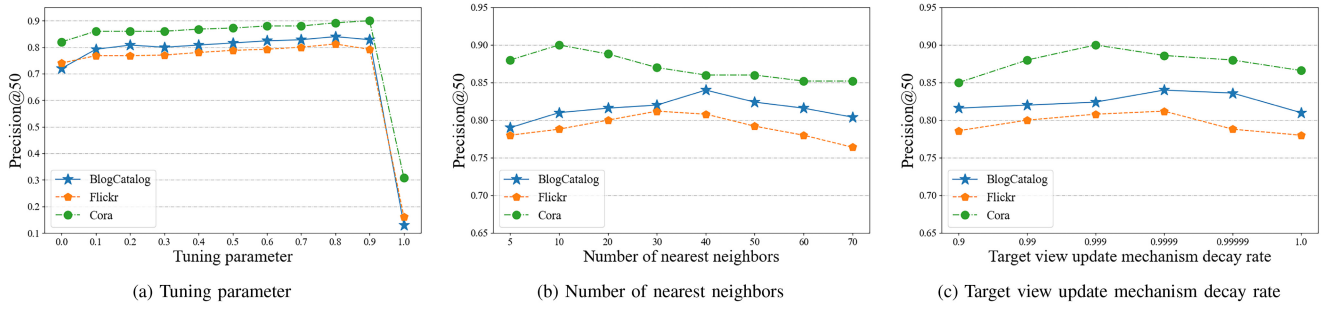


Fig. 3. Experimental results for parameter analysis. (a), (b), and (c) show the impact of different tuning parameter, numbers of nearest neighbors, and target view update mechanism decay rate w.r.t. Precision@50, respectively.

TABLE V
EFFECT OF DIFFERENT EDGE ATTRIBUTES W.R.T. PRECISION@K

Datasets	Precision@K											
	BlogCatalog				Flickr				Cora			
K	50	100	200	300	50	100	200	300	50	100	200	300
$d(v_i)$	0.812	0.730	0.591	0.480	0.788	0.780	0.677	0.582	0.854	0.770	0.668	0.460
$d(N(v_i))$	0.820	0.734	0.594	0.482	0.796	0.786	0.681	0.592	0.858	0.776	0.672	0.464
$CN(v_i, v_j)$	0.832	0.748	0.603	0.487	0.812	0.800	0.693	0.607	0.892	0.798	0.686	0.480
$Adamic(v_i, v_j)$	0.836	0.754	0.605	0.490	0.808	0.792	0.690	0.602	0.874	0.790	0.685	0.475
$Jaccard(v_i, v_j)$	0.832	0.750	0.600	0.486	0.800	0.790	0.686	0.596	0.870	0.780	0.680	0.472
$PA(v_i, v_j)$	0.828	0.744	0.598	0.484	0.804	0.792	0.687	0.600	0.870	0.784	0.677	0.470
Mean of six indices	0.840	0.758	0.612	0.493	0.810	0.796	0.690	0.605	0.900	0.802	0.693	0.483

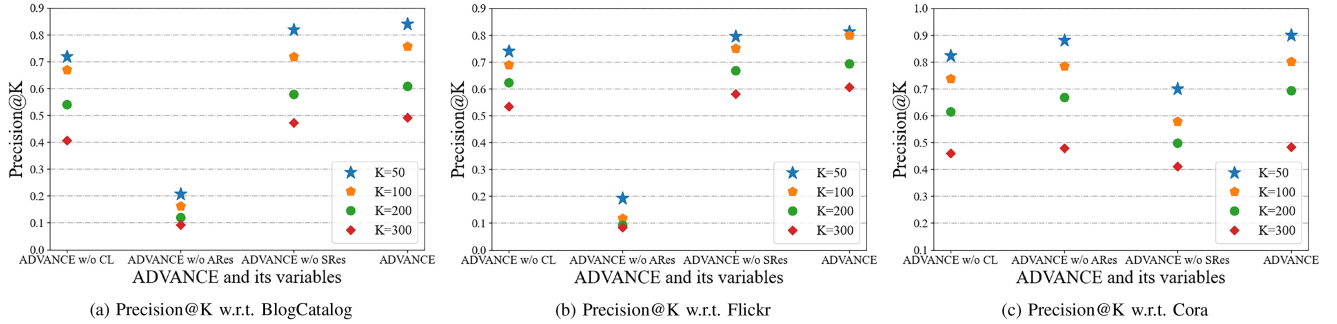


Fig. 4. Experimental results for ablation study. (a), (b), and (c) show the performance of ADVANCE and its variants w.r.t. Precision@K.

The number of nearest neighbors k is crucial for constructing the self-enhanced view, which reflects the neighborhood size of each node in the self-enhanced view. Smaller values of k may not enable nodes to aggregate neighborhood features adequately, while larger values of k may add invisible noise to node attributes aggregation. Fig. 3(b) shows the performance of ADVANCE with different numbers of nearest neighbors. Overall, for BlogCatalog, Flickr, and Cora datasets, the best performances are achieved when the number of nearest neighbors k are set as 40, 30, and 10, respectively.

The target view update mechanism decay rate δ plays a critical role in controlling the update process of the target view. A larger decay rate will cause the target view to be updated too slowly, while a smaller decay rate may cause the target view to update too sharply. When the decay rate $\delta = 1$, the target view is never updated and maintains the original structure. The selection of the decay rate for the target view update mechanism is critical in determining whether the target view can effectively alleviate the structural anomalies and guide the self-enhanced view to learn more valuable

network information. Fig. 3(c) shows the performance of the ADVANCE framework for different target view update mechanism decay rates. In the experiment, we set the target view update mechanism interval ε to 0 for all three datasets. As can be observed, various networks require distinct rates of update for the target view. The optimal decay rate is $\delta = 0.9999$ for BlogCatalog dataset and Flickr dataset and $\delta = 0.999$ for the Cora dataset. Furthermore, the performance of ADVANCE decreases when the decay rate $\delta = 1$, which proves the effectiveness of the target view update mechanism.

Edge attributes represent the edge attributes information used when initializing the target view. Section IV-A1 mentions that six similarity indices are considered to calculate the similarity of two connected nodes as edge attributes. The experimental results are shown in TABLE V, where *Mean of six indices* denotes the average of the calculated results of six similarity indices as the edge attributes. It can be seen that different networks have different adaptability to edge attributes. For BlogCatalog and Cora datasets, it is more advantageous to consider six edge attributes simultaneously,

while for Flickr dataset, using common neighbors to calculate edge attributes is more effective in detecting anomalous nodes.

F. Ablation Study (RQ3)

We further compare the results of ADVANCE and its variables, namely ADVANCE w/o CL, ADVANCE w/o ARes and ADVANCE w/o SRes. ADVANCE w/o CL, ADVANCE w/o ARes, and ADVANCE w/o SRes denote the framework without contrastive learning component (i.e., μ is set to 0), without attribute reconstruction component (i.e., α is set to 0) and without structure reconstruction component (i.e., α is set to 1) in anomaly detection, respectively. As we can see in Fig. 4, ADVANCE outperforms all the variants consistently on all three datasets, which indicates that all three components in ADVANCE contribute to the detection of anomalies. Especially, ADVANCE outperforms ADVANCE w/o CL by 12%, 13.2%, 11.6% on three datasets w.r.t. Precision@50, respectively, which demonstrates the contrastive-based module is crucially important to ADVANCE. The attribute reconstruction component and structure reconstruction component reconstruct the node attributes and topology structure of the original graph, respectively, so as to identify the anomalies in the network from multiple perspectives. According to the experimental results, the importance of attribute reconstruction and structure reconstruction in detecting anomalies is different for different attributed networks. For BlogCatalog and Flickr datasets, the node attributes contribute more to finding anomalous nodes, while for Cora dataset, paying attention to topology structure is more conducive to detecting anomalies in attributed networks.

VI. CONCLUSION

In this paper, we propose a novel anomaly detection framework based on graph contrastive learning, namely ADVANCE, which is consisted of two modules: graph contrastive learning module and network reconstruction module. ADVANCE innovatively constructs the target view and learnable self-enhanced view and takes advantage of the graph contrastive learning method to maximize the consistency between the target view and learnable view, which captures network information related to anomalies without the guidance of external information (i.e., labels). Next, the node attributes and topology structure of the original attributed networks are reconstructed by the network reconstruction module. Finally, two complementary modules are jointly trained to achieve more effective anomaly detection. A series of experiments on three benchmark datasets prove the effectiveness of ADVANCE. The proposed ADVANCE framework facilitates the early detection of fraudulent users and illicit financial transaction activities within consumer electronic networks, thereby furnishing consumer electronics with a dependable security assurance.

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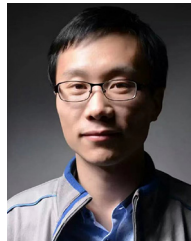
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Bo Xu (Member, IEEE) received the B.Sc. and Ph.D. degrees from the Dalian University of Technology, China, in 2007 and 2014, respectively, where she is currently an Associate Professor with the School of Software. Her current research interests include social computing, data mining, information retrieval, and natural language processing.



Jinpeng Wang received the B.Sc. degree in software engineering from the Dalian University of Technology, China, in 2021, where he is currently working the master's degree with the School of Software. His research interests include data science, social computing, and graph learning.



Zhehuan Zhao (Member, IEEE) received the B.Sc. and Ph.D. degrees from the Dalian University of Technology, China, in 2010 and 2017, respectively, where he is currently an Associate Professor with the School of Software. His current research interests include knowledge graph, data mining, and natural language processing.



Hongfei Lin received the M.S. degree from the Dalian University of Technology, China, in 1992, and the Ph.D. degree from Northeastern University, China, in 2000. He is currently a Professor with the College of Computer Science and Technology, Dalian University of Technology. He has published over 100 research papers in various journals, conferences, and books. His research interest includes text mining for biomedical literatures, biomedical hypothesis generation, information extraction from huge biomedical resources, sentimental analysis, and opinion mining.



Feng Xia (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees from Zhejiang University, Hangzhou, China. He is a Professor with the School of Computing Technologies, RMIT University, Australia. He has published over 300 scientific papers in international journals and conferences (such as IEEE TAI, TKDE, TNNLS, TC, TMC, TPDS, TBD, TCSS, TNSE, TETCI, TETC, THMS, TVT, TITS, TASE, ACM TKDD, TIST, TWEB, TOMM, WWW, AAAI, SIGIR, WSDM, CIKM, JCDL, EMNLP, and INFOCOM). His research interests include data science, artificial intelligence, graph learning, digital health, and systems engineering. He is a Senior Member of ACM, and an ACM Distinguished Speaker.